

Assignment 2: Coding Basics

Josh Salzberg

OVERVIEW

This exercise accompanies the lessons/labs in Environmental Data Analytics on coding basics.

Directions

1. Rename this file `<FirstLast>_A02_CodingBasics.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.
6. After Knitting, submit the completed exercise (PDF file) to Canvas.

Basics, Part 1

1. Generate a sequence of numbers from one to 55, increasing by fives. Assign this sequence a name.
2. Compute the mean and median of this sequence.
3. Ask R to determine whether the mean is greater than the median.
4. Insert comments in your code to describe what you are doing.

```
#1.  
fiveseq <- seq(1,55,5) # generate sequence from 1 to 55, increasing by fives; stores in variable fiveseq  
#2.  
mean(fiveseq) # return the mean of fiveseq
```

```
## [1] 26
```

```
median(fiveseq) #return the median of fiveseq
```

```
## [1] 26
```

```
#3.  
mean(fiveseq) > median(fiveseq) # return Boolean evaluation of whether the mean is greater than the med
```

```
## [1] FALSE
```

Basics, Part 2

5. Create three vectors, each with four components, consisting of (a) student names, (b) test scores, and (c) whether they are on scholarship or not (TRUE or FALSE).
6. Label each vector with a comment on what type of vector it is.
7. Combine each of the vectors into a data frame. Assign the data frame an informative name.
8. Label the columns of your data frame with informative titles.

```
student <- c("Charlie Buckets", "Samwise Gamgee", "Clark Kent", "Lark Trent") # list of our students
score <- c(75, 84, 33, 101) # Our students' latest test scores
scholar <- c(TRUE, FALSE, FALSE, FALSE) #Students' Scholarship Status
schoolrecords <- data.frame("Student Name" = student, "Latest Test Result" = score, "On Scholarship?" =
```

9. QUESTION: How is this data frame different from a matrix?

Answer: A dataframe allows for multiple types of data across columns, whereas a matrix requires all the data to be the same type.

10. Create a function with one input. In this function, use `if...else` to evaluate the value of the input: if it is greater than 50, print the word “Pass”; otherwise print the word “Fail”.
11. Create a second function that does the exact same thing as the previous one but uses `ifelse()` instead of `if...else`.
12. Run both functions using the value 52.5 as the input
13. Run both functions using the **vector** of student test scores you created as the input. (Only one will work properly...)

```
#10. Create a function using if...else
iffunction <- function(uservalue){
  if(uservalue > 50){
    print("Pass")}
  else {
    "Fail"
  }
}

#11. Create a function using ifelse()
ifelsefunction <- function(x){
  ifelse(x>50, "Pass", "Fail")
}

#12a. Run the first function with the value 52.5
iffunction(52.5)
```

```
## [1] "Pass"
```

```
#12b. Run the second function with the value 52.5
ifelsefunction(52.5)
```

```
## [1] "Pass"
```

```
#13a. Run the first function with the vector of test scores  
#iffunction(score)  
#13b. Run the second function with the vector of test scores  
ifelsefunction(score)
```

```
## [1] "Pass" "Pass" "Fail" "Pass"
```

14. QUESTION: Which option of `if...else` vs. `ifelse` worked? Why? (Hint: search the web for “R vectorization”)

Answer: The ‘ifelse’ function worked because it has vectorization, which allows the command to run the code on each element of the vector. ‘if...else’ works only with a loop.

NOTE Before knitting, you’ll need to comment out the call to the function in Q13 that does not work. (A document can’t knit if the code it contains causes an error!)